Sayan Ghosh

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SUMMARY

Fresher AI + Backend Developer with hands-on experience building AI-powered applications and backend systems using Python, FastAPI, AWS, Terraform, RAG, LangChain/LangGraph, and vector databases. Experienced in designing API-driven architectures, implementing authentication with AWS Cognito and API Gateway, and building cloud-based workflows using Lambda, S3, DynamoDB, and SQS. Strong interest in scalable backend systems, AI applications, and cloud engineering.

EDUCATION

JIS School of Polytechnic
Diploma inComputerScience& Technology *2023–2026*

JIS Collage of Engineering
B-techinComputerScince&Technology(AI and ML) *2026–2029*

PROJECTS

Nexsus — AI Backend & Cloud Platform (developement) | Python | AWS | Terraform | Lambda | API Gateway | Cognito | S3 | DynamoDB

- •Designed a cloud-based backend architecture using AWS Lambda, API Gateway, S3, DynamoDB, and IAM.
- •Implemented Cognito-based authentication and API authorization using JWT tokens.
- •Worked with API Gateway authorizers to securely pass authenticated user identity to backend services.
- •Used Terraform to provision and manage AWS infrastructure as code.
- Designed API-driven workflows for AI processing, file storage, and asynchronous tasks using SQS.

AI Document RAG System(Prototype) | Python | FastAPI | LangChain | LlamaIndex | FAISS/Qdrant | Hugging Face

- Built a document-processing and Retrieval-Augmented Generation (RAG) pipeline.
- Implemented PDF ingestion, document chunking, embeddings, vector storage, and retrieval.
- •Worked with FAISS and Qdrant for semantic/vector search.
- Used Hugging Face embedding models and LLM-based pipelines for AI-powered document querying.
- Designed workflow orchestration using LangGraph, including retry and failure-handling logic.

AI Video Generation Backend (⚠ not deployed) | Python | AWS | S3 | SQS | Lambda | Hugging Face | Streamlit

- •Designed an AI video-generation workflow with a preview → approval → queue → generation pipeline.
- Used Amazon SQS for asynchronous video-generation jobs.
- Used Amazon S3 for temporary and generated video storage.
- Implemented presigned URLs for controlled access to stored files.
- Designed backend responses and error-handling patterns for AI processing services.

Medical AI Chatbot (Prototype) | Python | FastAPI | LangChain | RAG | Streamlit | Hugging Face

- Developed a medical-focused AI chatbot using RAG and LLMs.
- Built a Streamlit chat interface and FastAPI backend.
- Implemented document retrieval and contextual question answering.
- •Worked with embeddings, vector search, conversation state, and external information retrieval.

TECHNICAL SKILLS

Languages: Python, Bash,json scripiting

Backend: FastAPI, REST APIs, HTTPX
AI/ML: LangChain, LangGraph, RAG, LlamaIndex, Hugging Face, Embeddings, FAISS, Qdrant
Cloud & AWS: Lambda, API Gateway, Cognito, S3, DynamoDB, SQS, IAM, CloudWatch
Infrastructure: Terraform, Docker, Docker Compose
Databases: PostgreSQL, DynamoDB, Redis, Vector Databases
Messaging: Apache Kafka, Amazon SQS
Tools: Git, GitHub, Streamlit, Prometheus, Grafana
Other: JWT, OAuth 2.0, Presigned URLs, Microservices
Skills: CICD, Terraform, Docker

KEY STRENGTHS

- AI Application Development - Backend API Development - AWS Cloud Architecture - Infrastructure as Code with Terraform - RAG & LLM Applications - Authentication & Authorization - Asynchronous / Event-Driven Systems - Problem Solving & Debugging - systems design (still learning)